

CCNA 200-301 - Chapter 26: Network Address Translation (NAT)

What is NAT?

NAT translates Private IP addresses into Public IP addresses so devices inside a LAN can access the Internet.

Why NAT?

- IPv4 addresses are limited.
- The Internet understands Public IP addresses only.
- NAT is a temporary solution until IPv6.

How NAT Works

The router replaces the Source Private IP with a Public IP before sending traffic to the Internet, then translates the reply back to the original device.

NAT Types

Static NAT

- One-to-One mapping.
- Each device has its own Public IP.
- Suitable for servers.
- Consumes many Public IPs.

Dynamic NAT

- Uses a Pool of Public IPs.
- Assigns any available Public IP temporarily.
- IP returns to the pool after the session ends.
- No free Public IP = no new connections.

PAT (NAT Overload)

- Most common type.
- Many devices share one Public IP.
- Uses Source Port numbers to distinguish connections.
- Saves Public IP addresses.

Packet Fields

Field	Description
Source IP	Translated to Public IP
Destination IP	Normally unchanged

Source Port	Used by PAT
Destination Port	Destination service port

Common Ports

Port	Service
20/21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
67/68	DHCP
80	HTTP
443	HTTPS

Comparison

Feature	Static	Dynamic	PAT
Mapping	One-to-One	Pool	Many-to-One
Public IP	One/device	Pool	One/Few
Ports	No	No	Yes
Usage	Servers	Limited	Most Common

Exam Notes

- Internet uses Public IP addresses.
- Static NAT = One-to-One.
- Dynamic NAT = Pool.
- PAT = NAT Overload and is the most common.
- IPv6 is the permanent solution to IPv4 exhaustion.